Adaptive influence diagnostics in regression models with highly correlated designs

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Abstract

In models with highly correlated designs, existing influence diagnostic tools tend to suffer from false flagging of normal observations, which often leads to misleading inference. To overcome this challenge, an adaptive Cook's distance (ACD) for diagnosing influential observations in both low- and high-dimensional single-index models in the presence of multicollinearity and outlier contamination is proposed. ACD is a model-free technique built on sparse local linear gradients to temper leverage effects. In simulations spanning low- and high-dimensional design settings with moderate and strong correlations, ACD based on LASSO (ACD-LASSO) and SCAD (ACD-SCAD) penalties reduced masking and swamping relative to classical Cook's distance and local influence as well as the recently proposed DF-Model and Case-Weight adjusted solution for LASSO regression. Trimming points flagged by ACD stabilizes variable selection while preserving core signals. Applications to two datasets - Monocyte Chemoattractant Protein-1 and a high-dimensional riboflavin genomics experiment show consistent gains in selection stability and interpretability.

Keywords: Outlier detection, multicollinearity, single–index models, sparse local gradients, local influence.

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1 Introduction

Effective regression analysis requires rigorous diagnostics to assess the adequacy of the assumed model and the stability of estimates, which is difficult to achieve when *influential observations* are present in the data. An observation is considered influential if its inclusion or omission significantly alters the regression estimates (Cook, 1977). Thus, a small fraction of such atypical data points can dramatically result in misleading inference including variable selection and prediction. This has motivated the development of diagnostic tools (Cook, 1977; Belsley et al., 1980; Prendergast, 2008; Soale, 2025).

Existing influence diagnostics can be broadly categorized into two: the *case-deletion* approach and the *perturbation* method. The case-deletion technique involves quantifying the effect of omitting one or more observations on the regression estimates, which dates back to Cook's distance, DFBETAS, and DFFITS (Cook, 1977; Cook and Weisberg, 1980; Chatterjee and Hadi, 1986). *Single-case deletion* is shown to suffer from masking (failure to detect a true influential point) and swamping (false flagging a normal point as influential) due to adjacent outliers, especially, when the proportion of influential observations in the data is substantial (Belsley et al., 1980). Thus, the *multiple-case deletion* was proposed, see Belsley et al. (1980); Peña (2005); Peña and Yohai (1999); Nurunnabi et al. (2014); Roberts et al. (2015), to name a few. The perturbation method On the other hand, involves adding minor perturbations to the model, usually in the predictor or the response, to assess its effect on the regression estimates. Earlier works in this directions can be traced to (Hodges and Moore, 1972; Belsley et al., 1980; Polasek, 1984; Cook, 1986).

Medical and biological data typically involve highly correlated predictors, especially

in the form of interaction terms, which exacerbate the influence of unusual observations. To handle high-dimensionality, sparsity–inducing diagnostics such as the difference in the selected model (DF-Model; Rajaratnam et al. (2019)) and Case-Weight adjusted solution for LASSO (CW-LASSO; Jiao and Lee (2025)) were proposed. Both techniques are based on the ℓ_1 regularization (Tibshirani, 1996), which is shown to be highly sensitive to unusual observations in terms of both coefficient magnitudes and model selection (Kim et al., 2015; Rajaratnam et al., 2019). Other techniques based on marginal correlation have been proposed for high dimensionality (Zhao et al., 2013; Wang et al., 2019) based on sure independence screening (Fan and Lv, 2008) and least trimmed squares (Rousseeuw and Leroy, 2003).

While these aforementioned techniques work under their specific model formulation, they are prone to model misspecification and less reliable in practice as the true model is often unknown or usually more complex in real medical applications. As noted by Cook (1986), the local influence methods rely on a "well-behaved" likelihood, which is not always guaranteed. Additionally, Wang et al. (2019)'s proposal relies on obtaining a "clean subset" of the data based on the screening in the first step, which may suffer from selection error. The LASSO-based techniques of Rajaratnam et al. (2019) and Jiao and Lee (2025) also inherit the disadvantages associated with LASSO such as the grouping effect problem when the predictors are highly correlated. Moreover, these techniques are handicapped by severe collinearity in the predictors, which exacerbates the masking and swamping problem regardless of the dimensionality.

In this paper, we propose a flexible model-free extension of the classical Cook's distance

method called the *Adaptive Cook's Distance (ACD)*, which is applicable in both low- and high-dimensional settings. Our adaptive Cook's distance method does not require specifying a response link as it utilizes sparse local linear gradients, which makes it suitable for general single index models. ACD exploits the connection between case-deletion and local linear approximation to stabilize influence assessment under collinearity and contamination. Kernel weights are employed to dampen the effect of leverage points while flexible sparsity-inducing penalties are incorporated to deal with multicollinearity via variable selection in the local estimation. ACD accommodates various sparsity-inducing penalties including LASSO, SCAD (Fan and Li, 2001), MCP (Zhang, 2010), and others. We estimate the effective direction as the leading principal component of the sparse local gradients, which is then used to compute the adaptive Cook's distance measure.

The remainder of this article is organized as follows. Section 2 proposes the model and index estimation via local linear regression. In Section 3, we establish the connection between local linear regression and the case-deletion technique, and proceed to introduce the adaptive Cook's distance. Section 4 reports extensive simulation studies under various contamination and collinearity scenarios, followed by two real–data applications in Section 5. Concluding remarks are given in Section 6, with technical details and proofs relegated to the Appendix.

1.1 A Motivating Example

We illustrate the adaptive Cook's distance using a classical linear regression model contaminated by influential observations under varying correlation structures in the design

matrix. Specifically, we set $(n, p) = (100, 10)$ and take the regression coefficient vector as $\beta = (3, 1.5, 0, 0, 2, 0, \ldots, 0)^\top \in \mathbb{R}^p$. Of the 100 observations, 95 are generated according to

$$\boldsymbol{x}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad y_i = 0.5 + \boldsymbol{x}_i^\top \beta + 0.8\varepsilon_{1i}, \quad \varepsilon_{1i} \sim N(0, 1),$$

representing the uncontaminated sample. The remaining 5 observations are generated from a heavy-tailed distribution with shifted mean,

$$\boldsymbol{x}_i \sim t_{10}(\mathbf{5}, \boldsymbol{\Sigma}), \qquad y_i = 0.5 + \boldsymbol{x}_i^\top \beta + \varepsilon_{2i}, \quad \varepsilon_{2i} \sim \chi^2(5),$$

to mimic influential cases. Two covariance structures for $\boldsymbol{\Sigma}$ are considered:

(a) independence, $\boldsymbol{\Sigma} = \mathbf{I}_p$,

(b) autoregressive correlation, $\boldsymbol{\Sigma}_{kl} = 0.5^{|k-l|}$, $k, l = 1, \ldots, p$.

A scatter plot illustrating the uncontaminated and contaminated samples is displayed in Figure 1.

Notice that although the regression coefficients and the set of outliers are the same across both designs, the relative performance of the influence diagnostics differ markedly (Figures 2–3). Under the independent design, the classical Cook's distance (Cook, 1977) and local influence (Cook, 1986) correctly identified most of the outliers, with the exception of observation 83. The DF-model (Rajaratnam et al., 2019) captured all outliers but suffers from swamping while the case-weight adjusted solution for LASSO (Jiao and Lee, 2025)'s could not capture observation 16. In contrast, our adaptive Cook's distance with LASSO penalty accurately detected the full set of outliers without false positives, and all but observation 19 with the SCAD penalty.

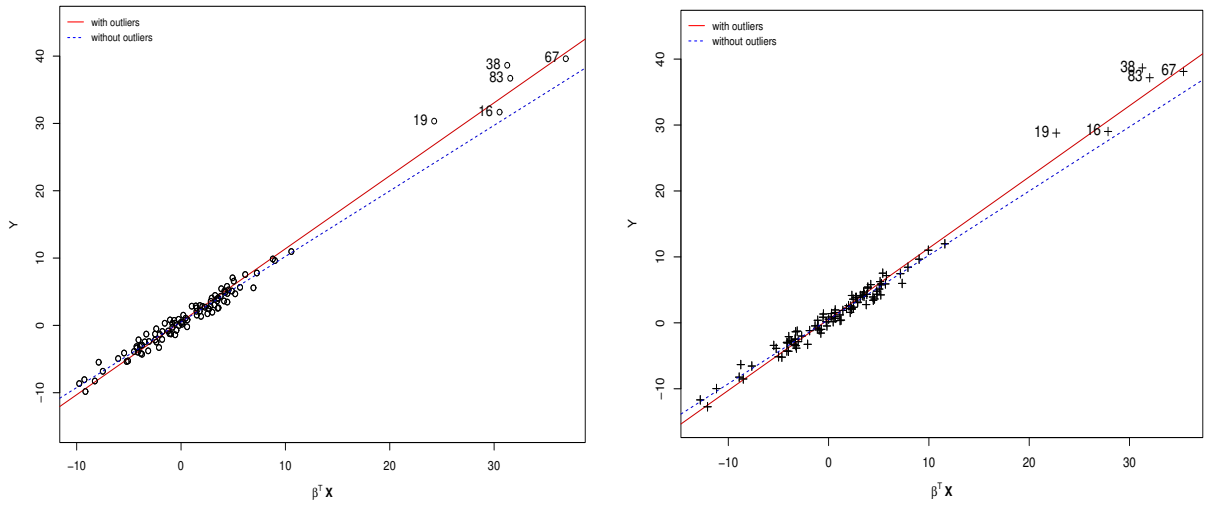

Figure 1: Scatter plot for design with independent correlation (left) and autoregressive correlation (right), with labels denoting the outliers. Blue line indicates the best fit line in the uncontaminated sample while the red line indicates the shift due to the outliers

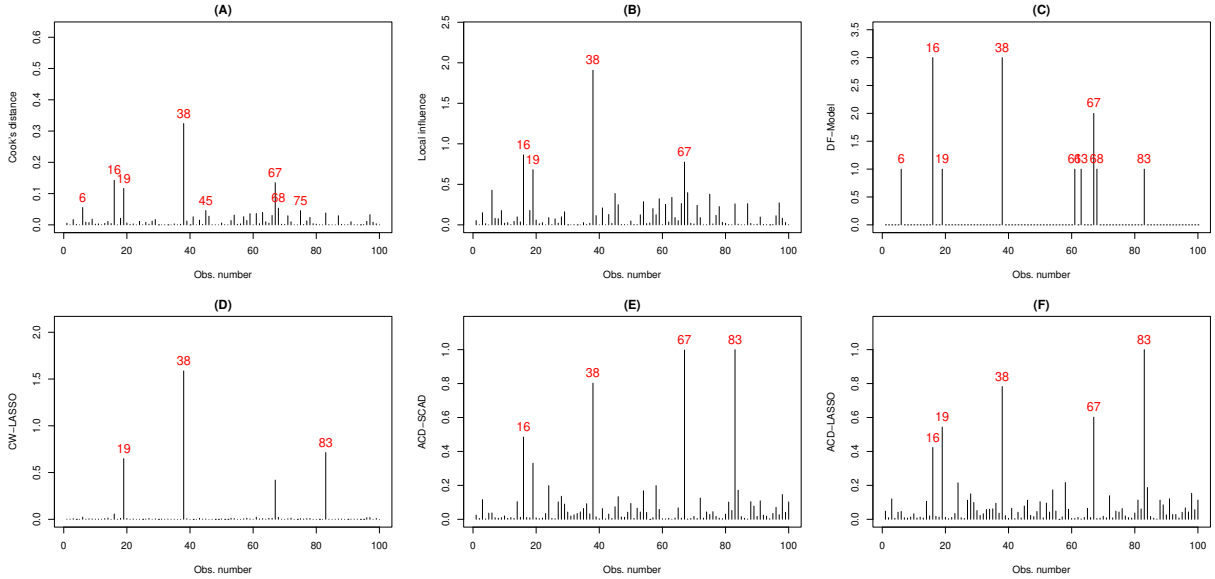

Figure 2: Influence measures under independent design

In the correlated design, Cook's distance and the DF-model become less reliable as both struggle with swamping. The local influence and case-weight for LASSO responded in the opposite direction, i.e., struggle with masking. In comparison, the adaptive Cook's distance remained stable and accurate, consistently detecting only the true outliers. Increasing the correlation to 0.7 and 0.9 further amplified the shortcomings of the existing methods. In all cases, the adaptive Cook's distance retained its robustness, providing reliable identification of influential cases even in the presence of severe multicollinearity. In the sections that follow, we will describe the development of the adaptive Cook's distance.

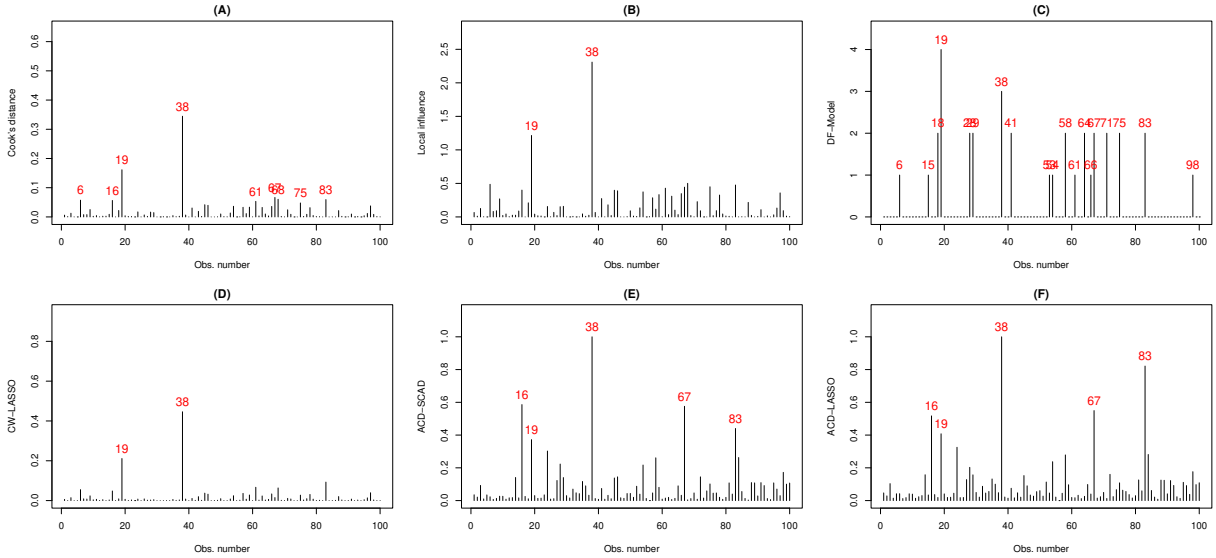

Figure 3: Influence measures under correlated design

2 Model

For a scalar response $Y \in \mathbb{R}$ and a p-dimensional predictor $\mathbf{X} \in \mathbb{R}^p$, consider the single-index model:

$$Y = f(\boldsymbol{\beta}^\top \mathbf{X}) + \epsilon, \tag{1}$$

where the function $f(.)$ is unknown, the parameter $\boldsymbol{\beta} \in \mathbb{R}^p$ with $\|\boldsymbol{\beta}\| = 1$, the noise ϵ is such that $\mathrm{E}(\epsilon|\mathbf{X}) = 0$, and $\|.\|$ denotes the ℓ_2-norm. In generalized linear models, we assume $f(.)$ is known, which can easily lead to model misspecification. Thus, here we avoid such restrictions.

By (1), we also have that $f(\boldsymbol{\beta}^\top \mathbf{X}) = \mathrm{E}(Y|\mathbf{X})$, which can be approximated using local linear regression. Assuming f is differentiable at $\mathbf{X} = \boldsymbol{x}_0$, the conditional mean at this point can be approximated using the first-order Taylor expansion:

$$\mathrm{E}(Y|\mathbf{X} = \boldsymbol{x}_0) \approx a + \mathbf{B}^\top (\mathbf{X} - \boldsymbol{x}_0), \tag{2}$$

where $a = f(\boldsymbol{\beta}^\top \boldsymbol{x}_0) \in \mathbb{R}$ and $\mathbf{B} = \left.\dfrac{\partial f(\boldsymbol{\beta}^\top \mathbf{X})}{\partial \mathbf{X}}\right|_{\mathbf{X}=\boldsymbol{x}_0} \in \mathbb{R}^p$.

Proposition 1. *Suppose f is differentiable. Then $\boldsymbol{\beta}$ is in the space spanned by the leading eigenvector of* $\mathrm{E}(\mathbf{B}\mathbf{B}^\top)$.

Proposition 1 follows directly from Lemma 1 of Xia et al. (2002). The outer-product of gradient technique is well-known in the sufficient dimension literature and can be seen in Xia (2007); Wang and Xia (2008); Wang and Yao (2012), to name a few.

To obtain the estimates for $(a, \mathbf{B})^\top$ at $\boldsymbol{x}_0$, we solve

$$\arg\min_{a,\mathbf{B}} \mathrm{E}[(Y - a - \mathbf{B}^\top (\mathbf{X} - \boldsymbol{x}_0))^2 w(\boldsymbol{x}_0)], \tag{3}$$

where $w(\boldsymbol{x}_0)$ denotes some weighting function. Usually $w(\boldsymbol{x}_0) = K_h((\mathbf{X} - \boldsymbol{x}_0)/h)$, where $K_h(.)$ denotes a kernel function with bandwidth h. While there are several options for $K_h(.)$, a popular choice is the Gaussian radial basis function given by $K_h(u) = \frac{1}{\sqrt{2\pi}}\exp(-u^2/2h)$. This kernel gives lower weights to points that are farther from $\boldsymbol{x}_0$, thus, dampening the effect of distant high leverage points on the local estimate. Moreover, by choosing an optimal radius, we can obtain results similar to multiple-case deletion.

3 Adaptive Cook's Distance

3.1 Single-case Deletion Revisited

Consider the classical linear model $Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon$, where $\mathrm{E}(\epsilon|\mathbf{X}) = 0$. Given a random sample $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$ of $(\mathbf{X}, Y)$, the ordinary least squares (OLS) estimator is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{X}_n^\top Y_n, \tag{4}$$

where $\mathbf{X}_n = (\mathbf{1}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top$, $Y_n = (y_1, \ldots, y_n)^\top$, and $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p)^\top$. It is well-known that $\hat{\boldsymbol{\beta}}$ is susceptible to both outliers and collinearity in $\mathbf{X}$. Thus, the presence of a few influential observations coupled with mild collinearity can lead to unstable estimates, and consequently, misleading inference (Kim et al., 2015; Rajaratnam et al., 2019).

To measure the influence of an individual observation, a popular technique is to use the single-case deletion approach to see if there is a significant change in $\hat{\boldsymbol{\beta}}$ or the predicted response $\hat{Y}_n = \hat{\boldsymbol{\beta}}^\top \mathbf{X}_n$. Suppose $(\mathbf{X}_{-i}, Y_{-i}) = (\mathbf{X}_n, Y_n) \backslash \{(\boldsymbol{x}_i, y_i)\}$, i.e., the sample without

the ith observation. Then the corresponding OLS estimator based on $(\mathbf{X}_{-i}, Y_{-i})$ is given by

$$\hat{\boldsymbol{\beta}}^{-i} = (\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1}\mathbf{X}_{-i}^\top Y_{-i} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}\boldsymbol{x}_i}{1-h_{ii}} e_i, \tag{5}$$

where $h_{ii} = \boldsymbol{x}_i^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1}\boldsymbol{x}_i$ is the ith diagonal element of $\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ (hat matrix) and $e_i = y_i - \hat{y}_i$ is the residual of the ith observation. Observations with high values of h_{ii} are called *leverage points*. From (5), it is obvious that the deviation of $\hat{\boldsymbol{\beta}}^{-i}$ from $\hat{\boldsymbol{\beta}}$ can come from leverage points, high $|e_i|$, *ill-conditioning*[1] of $\mathbf{X}_n^\top \mathbf{X}_n$, or a combination of these factors. Notice that as both $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}^{-i}$ depend on $(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$, severe multicollinearity in $\mathbf{X}_n$ can lead to ill-conditioning of $\mathbf{X}_n^\top \mathbf{X}_n$, making the inverse highly unstable. In such a scenario, a small change in the $\boldsymbol{x}_i$'s or y_i's can result in huge changes in $\hat{\boldsymbol{\beta}}$.

Consider the case of the classical Cook's distance (Cook, 1977), which for the ith observation is given by

$$D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{-i})^\top \mathbf{X}_n^\top \mathbf{X}_n (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{-i})}{(p+1)\hat{\sigma}^2}, \tag{6}$$

where $\hat{\boldsymbol{\beta}}^{-i}$ is the OLS estimate without the ith observation and the estimated error variance $\hat{\sigma}^2 = (n-p-1)^{-1}\|Y_n - \hat{\boldsymbol{\beta}}^\top \mathbf{X}_n\|^2$ is based on all observations. If $\mathbf{X}_n$ suffers from multicollinearity, we expect the performance of the Cook's distance to degrade. Sadly, the same is true for the local influence measures and the recent extensions such as the DF-Model and CW-LASSO as illustrated in the motivating example.

[1] A square matrix $\mathbf{A}$ is ill-conditioned if the solution to $\mathbf{A}x = b$ it is extremely sensitive to small changes in $\mathbf{A}$ or b, resulting in wildly different solutions.

3.2 Case-Deletion: A Special Case of Local Linear Regression

The case-deletion technique is a special case of the local linear regression. To see this, notice that the estimator $\hat{\boldsymbol{\beta}}^{-i}$ given in (5) can also be expressed as

$$\hat{\boldsymbol{\beta}}^{-i} = (\mathbf{X}_n^\top \mathbf{W}_i \mathbf{X}_n)^{-1} \mathbf{X}_n^\top \mathbf{W}_i Y_n, \tag{7}$$

where the weighting function $\mathbf{W}_i = diag(w_{jj})$ is a diagonal matrix of binary weights with $w_{jj} = 0$ if $j = i$ and 1 if $j \neq i$. Similarly, the multiple case deletion is equivalent to setting the weights to 0 for all the deleted observations and 1 otherwise.

If we generalize the binary weights $\mathbf{W}_i$ to some function $w(\boldsymbol{x}_i)$, $\hat{\boldsymbol{\beta}}^{-i}$ becomes the solution to the local linear regression in (3). By adopting the kernel weights, we can improve the case deletion estimator $\hat{\boldsymbol{\beta}}^{-i}$ by choosing the weight function to dampen the effect of distant leverage points.

3.3 Adaptive Cook's Distance

To overcome the limitation of the classical Cook's distance in very high-dimensional settings ($p > n$), we impose a sparsity-inducing penalty on $\mathbf{B}$. Thus, we solve the penalized local weighted least squares:

$$(\hat{a}_i, \hat{\mathbf{B}}_i)^\top = \arg\min_{a,\mathbf{B}} \sum_{j=1}^{n} \left[y_j - a - \mathbf{B}^\top (\boldsymbol{x}_j - \boldsymbol{x}_i) \right]^2 w(\boldsymbol{x}_i) + P_\lambda(\mathbf{B}), \tag{8}$$

where $P_\lambda(\mathbf{B})$ is a sparsity-inducing penalty term and the weight $w(\boldsymbol{x}_i)$ is given by

$$w(\boldsymbol{x}_i) = \frac{\exp(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|/\tau^2)}{\sum_{j=1}^{n} \exp(-\|\boldsymbol{x}_i - \boldsymbol{x}_j\|/\tau^2)}.$$

The penalty $P_\lambda(\mathbf{B})$ can be any of the popular sparsity-inducing functions. Here, we will only focus on LASSO and SCAD, which are given by

$$\begin{cases} P_\lambda(\mathbf{B}) = \lambda \sum_{k=1}^{p} |B_k| \text{ for LASSO,} \\ P_\lambda(B_k) = \lambda \int_0^{|B_k|} \min\left\{1, (\gamma - t/\lambda)_+/(\gamma - 1)\right\} dt \text{ for SCAD,} \end{cases} \tag{9}$$

where $(c)_+ = \max(0, c)$ with $\gamma > 2$, typically set around 3.7. LASSO is effective at performing variable selection but tends to give biased estimates when there is strong multicollinearity as it arbitrarily selects one of the variables in the correlated group (Zou and Hastie, 2005). On the other hand, SCAD is designed to reduce bias and achieves "oracle properties" asymptotically (Fan and Li, 2001).

Let $\hat{\boldsymbol{\eta}}_i = (\hat{a}_i, \hat{\mathbf{B}}_i)^\top$ and $\boldsymbol{\Lambda}_n = (\hat{\boldsymbol{\eta}}_1, \ldots, \hat{\boldsymbol{\eta}}_n)^\top \in \mathbb{R}^{n\times(p+1)}$ be the vector of sparse local parameter estimates. By the reduced singular value decomposition (SVD), if $\boldsymbol{\Lambda}_n$ has rank r, then $\boldsymbol{\Lambda}_n$ can be decomposed as $\boldsymbol{\Lambda}_n = \mathbf{UDV}^\top$, where $\mathbf{U} = (\mathbf{u}_1, \ldots, \mathbf{u}_r) \in \mathbb{R}^{n\times r}$ and $\mathbf{V}^\top = (\mathbf{v}_1, \ldots, \mathbf{v}_{p+1}) \in \mathbb{R}^{r\times(p+1)}$ are orthonormal vectors, and $\mathbf{D} = diag(d_1, \ldots, d_r) \in \mathbb{R}^{r\times r}$ with $d_1 \geq d_2, \ldots, \geq d_r > 0$. Thus, the columns of $\mathbf{V}$ are the principal component loadings of $\boldsymbol{\Lambda}_n^\top \boldsymbol{\Lambda}_n$. Following Proposition 1, we retain only $\mathbf{v}_1$, the eigenvector corresponding to the leading eigenvalue.

By replacing $\hat{\boldsymbol{\beta}}^{-i}$ with $\hat{\boldsymbol{\eta}}_i$ and $\boldsymbol{\beta}$ with $\mathbf{v}_1$, we define the adaptive Cook's distance for the ith observation as

$$D_i^A = \frac{(\mathbf{v}_1 - \hat{\boldsymbol{\eta}}_i)^\top \mathbf{X}_n^\top \mathbf{X}_n (\mathbf{v}_1 - \hat{\boldsymbol{\eta}}_i)}{(p+1)d_1^2}, \tag{10}$$

where the $\hat{\sigma}^2$ in (6) is replaced by the corresponding eigenvalue d_1^2 of $\mathbf{v}_1$. Note that because we are dealing with a squared distance, the lack of identifiability of $\mathbf{v}_1$ should not affect

D_i^A.[2]

Once, we compute $\mathbf{D}_n^A = (D_1^A, \ldots, D_n^A)^\top$, we proceed to scale the distances using the min-max normalization to obtain

$$\tilde{\mathbf{D}}_n^A = \frac{\mathbf{D}_n^A - \min(\mathbf{D}_n^A)}{\max(\mathbf{D}_n^A) - \min(\mathbf{D}_n^A)}, \tag{11}$$

with $0 \leq \tilde{D}_i^A \leq 1,\ i = 1, \ldots, n$. In the spirit of Rajaratnam et al. (2019), we consider observations with $\tilde{D}_i^A > mean(\tilde{\mathbf{D}}_n^A) + 2SD(\tilde{\mathbf{D}}_n^A)$ as influential. However, the investigator may choose a reasonable fixed cutoff, say, 0.5.

The adaptive Cook's distance is summarized in Algorithm 1. To ensure fair penalization, all the predictors are standardized to be on the same scale. Also, the algorithm relies on a number of tuning parameters, which need to be addressed in the implementation. First, the kernel weight in Step 2 depends on the parameter τ. To determine the optimal value, we follow the recommendation of Li et al. (2011) and compute it as

$$\hat{\tau} = \frac{1}{\binom{n}{2}} \sum_{i<j,\ j=2}^{n} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|.$$

Next, the regularization parameter λ in Step 4 is determined using the usual cross-validation over a grid of values. This can be done using standard packages such as *glmnet* or *ncvreg* in R statistical software.

[2] $\mathbf{v}_i$'s are only unique up to a sign flip.

Algorithm 1 Adaptive Cook's Distance

Input $(\mathbf{X}_n, Y_n)$

Marginally scale $\mathbf{X}_n$ as $\mathbf{Z}_n$ and center Y_n as Y_c

for $i = 1$ to n **do**

  Compute $\boldsymbol{w}_i \leftarrow \dfrac{\exp(-\|\boldsymbol{z}_i - \boldsymbol{z}_j\|/\hat{\tau}^2)}{\sum_{j=1}^{n} \exp(-\|\boldsymbol{z}_i - \boldsymbol{z}_j\|/\hat{\tau}^2)}, \; j = 1, \ldots, n$

  Set $\widetilde{Y}_n \leftarrow \sqrt{\boldsymbol{w}_i} Y_c$ and $\widetilde{\mathbf{Z}}_n \leftarrow \sqrt{\boldsymbol{w}_i}(\mathbf{Z}_n - \boldsymbol{z}_i)$

  Compute the LASSO/SCAD estimate $\hat{\boldsymbol{\eta}}_i = (\hat{a}_i, \hat{\mathbf{B}}_i)^\top$ for regressing $\widetilde{Y}_n$ on $\widetilde{\mathbf{Z}}_n$

end for

Set $\boldsymbol{\Lambda}_n \leftarrow (\hat{\boldsymbol{\eta}}_1, \ldots, \hat{\boldsymbol{\eta}}_n)^\top$ and a compute $(\hat{\mathbf{u}}_1, \hat{\mathbf{v}}_1, \hat{d}_1)^\top \leftarrow svd(\boldsymbol{\Lambda}_n)$

Compute $\tilde{\mathbf{D}}_n^A$

3.4 Extension to Non-monotone Link Functions

Because ACD is constructed from local (data–adaptive) gradients of the conditional mean, it is not restricted to the classical linear model or models with monotone link functions. It only requires that $f(\cdot)$ be at least first-order differentiable. It is worth noting that the classical linear model can also handle model misspecification but only when $f(.)$ is monotone and $\mathbf{X}$ follows an elliptical distribution (Li and Duan, 1989; Eaton, 1986).

To buttress this point, consider a slight modification of the motivating example:

$$Y \;=\; \left(0.5 + \boldsymbol{\beta}^\top \mathbf{X}\right)^2 + \varepsilon,$$

with the same settings as in the motivating example and $\boldsymbol{\Sigma} = \mathbf{I}_p$. The injected outliers remain $\{16, 19, 38, 67, 83\}$. Owing to the symmetry of the squared link, diagnostics that implicitly assume a linear link—Cook's distance, local influence, DF–Model and CW-LASSO

exhibit reduced sensitivity to these outliers (see Fig. 4). By contrast, ACD continues to prioritize the true outliers and avoids spurious flags, highlighting its robustness to model misspecification.

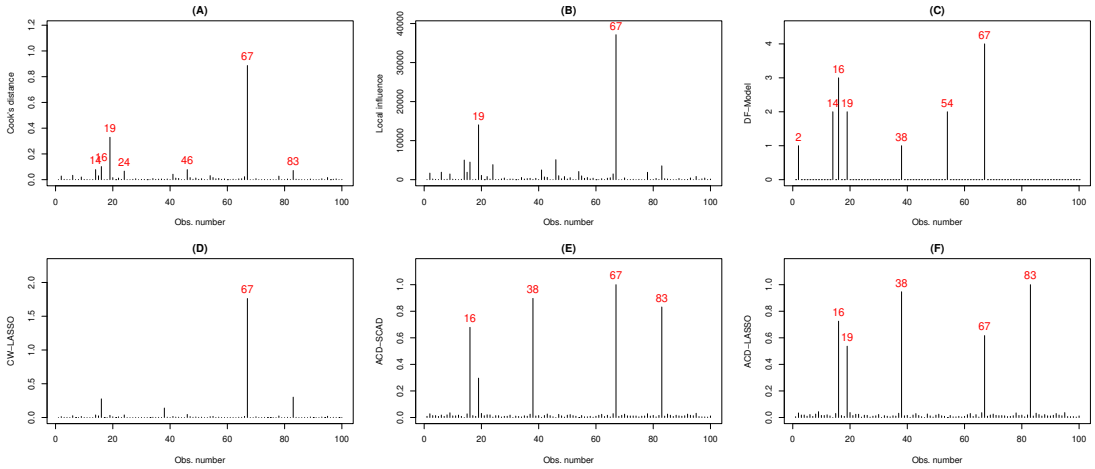

Figure 4: Detecting influential observations under symmetric link

4 Simulation Study

In this section, we first explore the performance of our proposal under the LASSO (ACD-LASSO) and SCAD (ACD-SCAD) penalties in diagnosing influential observations in synthetic data compared to the classical Cook's distance (CKD), local influence measure (LOC-INF), DF-Model, and Case-Weight adjusted method for LASSO (CW-LASSO). In the second part of the simulation study, we investigate the effect of trimming the flagged influential observations on variable selection based on the competing methods. In both studies, we consider low-dimensional ($p = 0.1n$), moderate-dimensional ($p = 0.5n$), and high-dimensional ($p = 2n$) designs under different correlation structures. For a fair comparison, we only focus on the linear model with identity link.

4.1 Detecting Influential Observations

Here, we consider two models. For each model we include $P_{out}\%$ of the observations as outliers. To evaluate performance, let $\mathcal{O} = \{\text{true outliers}\}$. We report the true positive rate of the detection as

$$\text{TPR} = \frac{|\{j \in \{1, \ldots, n\} : j \in \mathcal{O}\}|}{|\mathcal{O}|},$$

where j is an observation flagged as influential and $|.|$ denotes the cardinality of the set.

Model I: Set $n = 100$, $p = \{10, 50\}$, $\beta = (3, 1.5, 0, 0, 2, 0, \ldots, 0)^\top \in \mathbb{R}^p$ and $P_{out} = 5\%$. Next, generate 95 observations as $\boldsymbol{x}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $y_i = 0.5 + \beta^\top \boldsymbol{x}_i + 0.8\epsilon_{1i}$, where $\epsilon_1 \sim N(0, 1)$; and the remaining 5 observations as $\boldsymbol{x}_i \sim t_{10}(\mathbf{5}, \boldsymbol{\Sigma})$ and $y_i = 0.5 + \beta^\top \mathbf{X}_i + \epsilon_{2i}$, where $\epsilon_2 \sim \chi^2(5)$ with (a) $\boldsymbol{\Sigma} = 0.5^{|k-l|}$, and (b) $\boldsymbol{\Sigma}_{ij} = 1$ if $k = l$ and 0.5 otherwise, where $k, l = 1, \ldots, p$.

The box plots of the true positive rate for different influence detection measures under different design correlation structures are given in Figure 6. Under both design scenarios, we see that ACD based on SCAD and LASSO penalties yield the best results. The DF-Model is competitive but not very efficient followed closely by the classical Cook's distance (CKD). CW-LASSO and the local influence measure (LOC-INF) perform poorly with LOC-INF giving the worse performance. While all methods deteriorate under the exchangeable design in (b), we see that ACD remained consistent with the high performance compared to the other methods. When p increased, the performance of the classical Cook's distance deteriorated even more as expected.

Model II: We use the same data generation process as in Model 1 but set $(n, p) =$

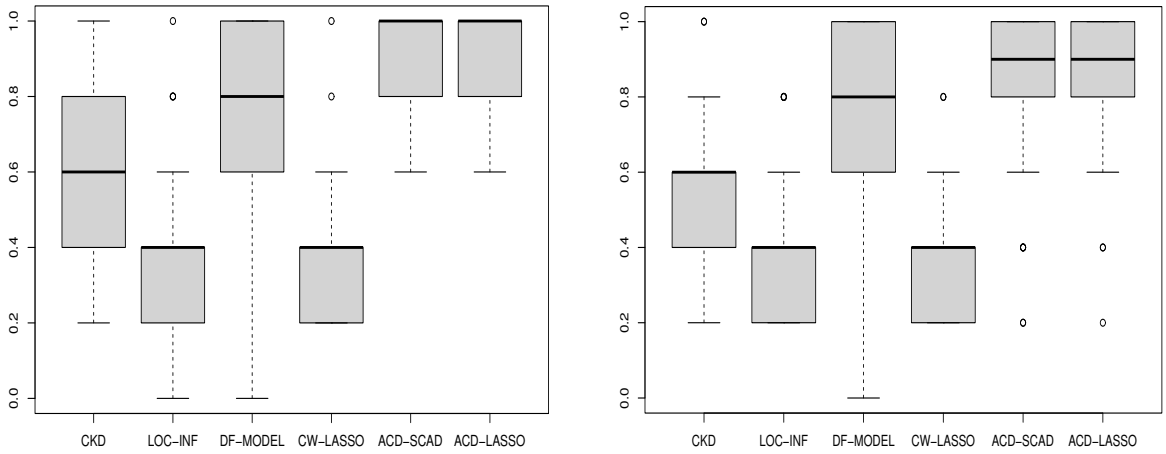

Figure 5: TPR of influence detection with $p = 10$ based on AR(1) (left) and exchangeable (right) design correlation with $\rho = 0.5$

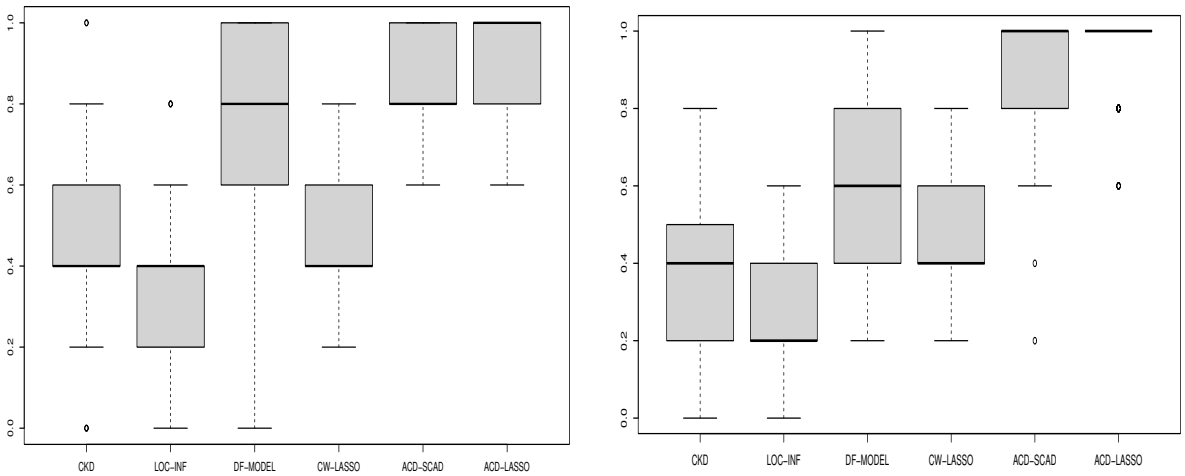

Figure 6: TPR of influence detection with $p = 50$ based on AR(1) (left) and exchangeable (right) design correlation with $\rho = 0.5$

$(100, 200)$, $P_{out} = 10\%$, and $\rho = 0.7$. Also, for $\boldsymbol{\beta}$, we allow only 5 active coefficients from the set $\{-2, -1, 0.5, 1.5, 3\}$, which are randomly assigned. Here, because $p >> n$, the classical

OLS does not apply. Hence, we leave out the classical Cook's distance and local influence measure. The results for 100 repeated samples are provided in Figure 7.

Again, in the high dimensions we see ACD under both LASSO and SCAD penalties outperforming the DF-Model and CW-LASSO. This demonstrates the flexibility and robustness of the adaptive Cook's distance in both low- and high-dimensional settings. All examples thus far, suggest that the choice of penalty term for the sparse local estimates does not have a substantial impact on the performance of the ACD.

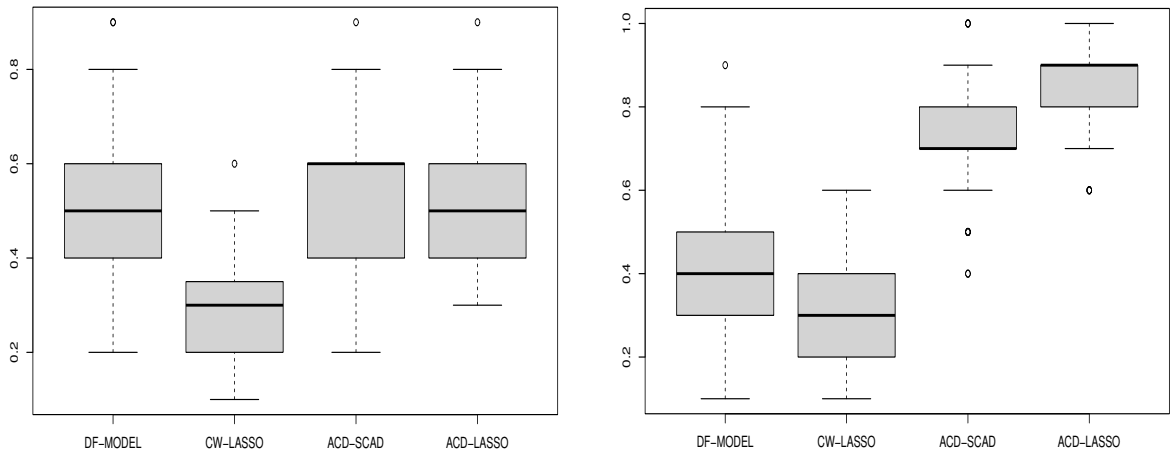

Figure 7: TPR of influence detection in high dimension based on AR(1) (left) and exchangeable (right) design correlation with $\rho = 0.7$

4.2 Impact of Trimming Influential Observations on Variable Selection

We study how removing observations flagged as influential affects the accuracy of the variable selection. To avoid LASSO's grouping behavior, the final selection is performed with SCAD regardless of the influence detector. The selection error is summarized by the false positive rate (FPR),

$$\mathrm{FPR} = \frac{\left|\{k \in \{1, \ldots, p\} : \ \widehat{\beta}_k \neq 0 \wedge \beta_k = 0\}\right|}{\left|\{k \in \{1, \ldots, p\} : \ \beta_k = 0\}\right|},$$

i.e., the proportion of truly null coefficients that are spuriously included. Alongside the estimator using all observations (ALL), we report results after trimming the cases identified by each detection method. Figures 9–10 display the distribution of FPRs across Monte Carlo replicates.

In Figure 9, for both AR(1) and exchangeable designs with $\rho = 0.5$, the analysis that retains all observations (ALL) and the one after trimming the influential observations flagged by Cooks distance (CKD) exhibit the highest false–positive rates (elevated medians with longer upper tails). Removing cases flagged by local influence (LOC-INF) lowers the rate of spurious inclusions, and trimming guided by LASSO diagnostics (DF-MODEL and CW-LASSO) yield additional reductions. The ACD-based trimming followed by SCAD consistently delivers the smallest and most stable FPRs, with boxplots concentrated near zero. The gap between ALL and trimmed analyses is slightly larger under the AR1 design, indicating greater sensitivity to influential cases when correlations are unevenly concentrated.

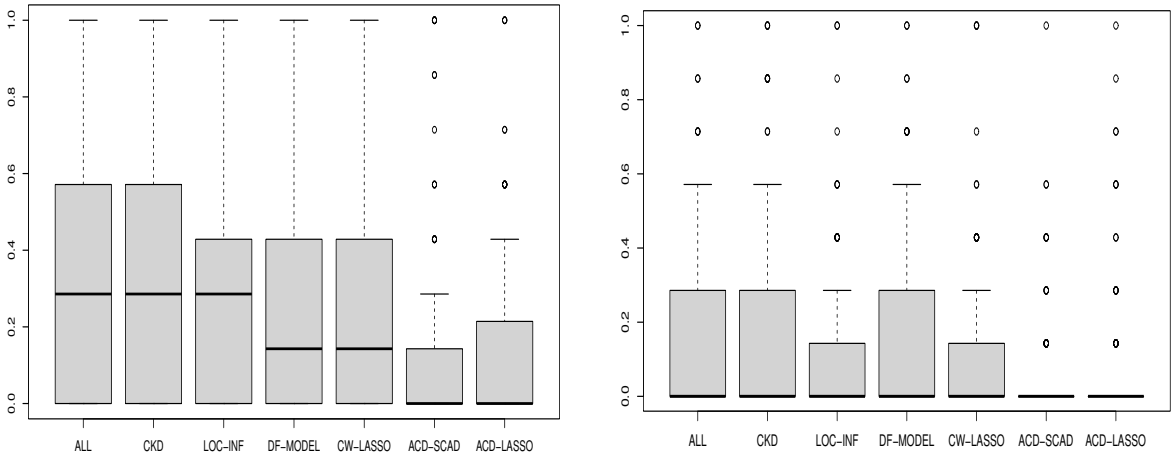

Figure 8: FPR with $p = 10$ based on AR(1) (left) and exchangeable (right) design correlations with $\rho = 0.5$ for Model I

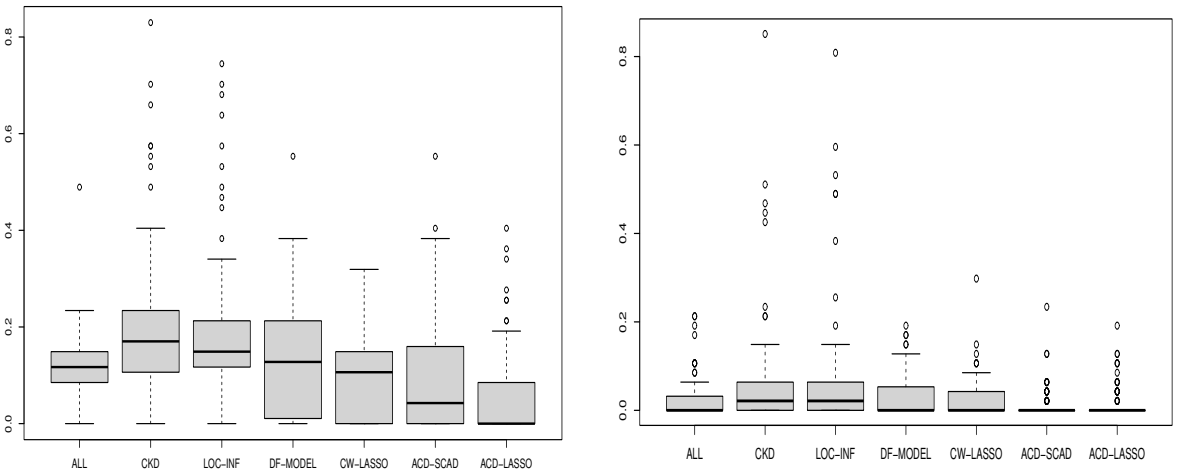

Figure 9: FPR with $p = 50$ based on AR(1) (left) and exchangeable (right) design correlations with $\rho = 0.5$ for Model I

Figure 10 considers a higher-correlation design ($\rho = 0.7$). Trimming remains beneficial: every influence detector reduces FPR relative to ALL, and the ACD+SCAD combination

again attains the lowest medians and the tightest dispersion. Although the absolute FPRs are small on this scale, the variance reductions are nontrivial and improve reproducibility across replicates.

Overall, we see that within the designs studied, a small number of influential observations can materially inflate false selections. Trimming flagged cases before SCAD selection generally reduces over-selection and stabilizes support recovery. The ACD-guided trimming provides the largest and most consistent FPR reductions while the model-based and coordinatewise detectors offer intermediate gains. For transparency, we recommend reporting results with and without trimming; when trimming is applied, an ACD-style screen followed by a SCAD refit gave the most stable selections in our experiments.

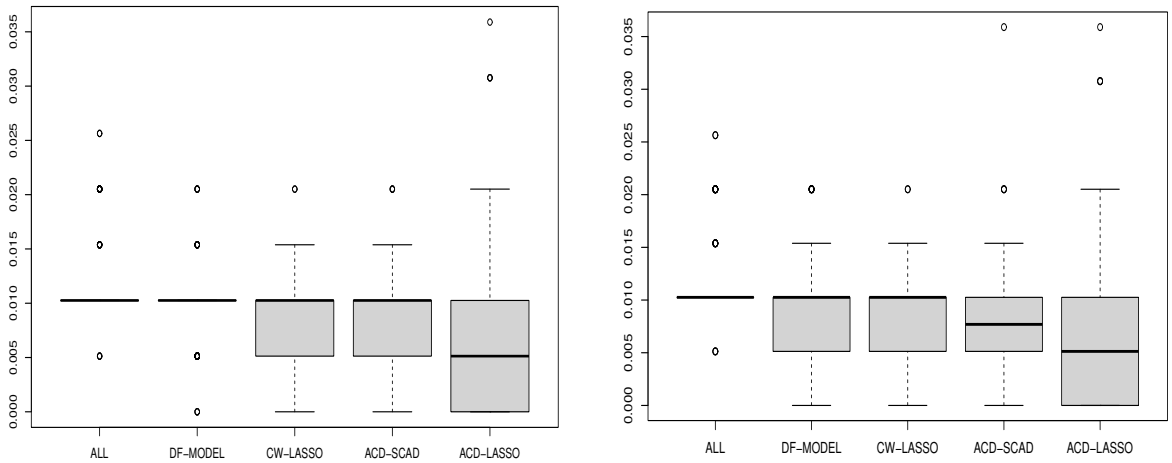

Figure 10: FPR in high dimensions based on AR(1) (left) and exchangeable (right) design correlations with $\rho = 0.7$

5 Real Applications

In this section we demonstrate the utility of the proposed adaptive Cook's distance in the analyses of Monocyte Chemoattractant Protein 1, which is plays a key role in immune response and cancer among others. We also apply it to the riboflavin gene expression dataset.

5.1 Application to Monocyte Chemoattractant Protein 1 (MCP-1) Data

Monocyte chemoattractant protein-1 (MCP-1) is a type of chemokine that regulate migration and infiltration of monocytes to tissues during inflammation (Deshmane et al., 2009). It is often overexpressed in breast cancer, particularly in more aggressive subtypes such as triple-negative breast cancer (TNBC) (Dutta et al., 2018). Thus, MCP-1 is one of the pro-metastatic chemokine that plays a critical role in breast cancer progression (Mulholland et al., 2023).

In this analysis, we explore the factors that influence the level of MCP-1 in serum based on a sample of 116 subjects including 64 breast cancer patients and 52 healthy controls. This dataset is publicly available on UCI Machine Learning Repository and details about the data collection can be found in (Patrício et al., 2018) [3]. We consider 9 clinical and demographic factors including age, body mass index (BMI), level of glucose, insulin, leptin, adiponectin, resistin, homeostasis model assessment (HOMA) index, and health status of the subject (breast cancer / healthy). The HOMA index is computed

[3] https://archive.ics.uci.edu/dataset/451/breast+cancer+coimbra

as the logarithm of fasting insulin level $\times$ fasting glucose level. Therefore, the HOMA index is highly correlated with insulin and glucose with correlation coefficients of 0.93 and 0.70, respectively. Also, as leptin is produced mainly by adipocytes, it is also positively correlated with BMI (correlation coefficient of 0.57), and there is also a positive correlation between glucose and insulin (correlation coefficient of 0.5).

Previous studies have found that the expression of MCP-1 in tumors varies across stages of breast cancer (Mulholland et al., 2023). Also, as seen in Figure 11, there is on significant difference in MCP-1 between those with breast cancer and the control. Therefore, we have to carefully appraise the influence the predictors on MCP-1, especially because of the collinearity in the predictors. In our sample, observations 38, 79, 88, and 89 have hat-values (h_{ii}) of at least 0.38, and the MCP-1 values range between 45pg/dL and 1700pg/dL with observations 41, 42, 55, 70, 114, 116 as outliers, see Figure 11. Thus, conventional influence summaries must be interpreted cautiously because leverage can be large even when the combined effect on fitted values is ambiguous.

Motivated by the collinearity in the predictors, we complement classical diagnostics with our robust detection scheme and a set of competitors. The results are provided in Figure 12. The classical Cook's distance, local influence, DF-Model, and CW-LASSO procedures largely agree on a small subset of cases, but the magnitude of their estimated influence depends on how each method mitigates collinearity. Overall, observations 38, 51, 79, and 89 appear to be the most influential based on these procedures. These observations also happen to be those with the highest leverage. On the other hand, the adaptive Cook's distance based on LASSO and SCAD both flagged observations 45, 47, 110, 111, and 116

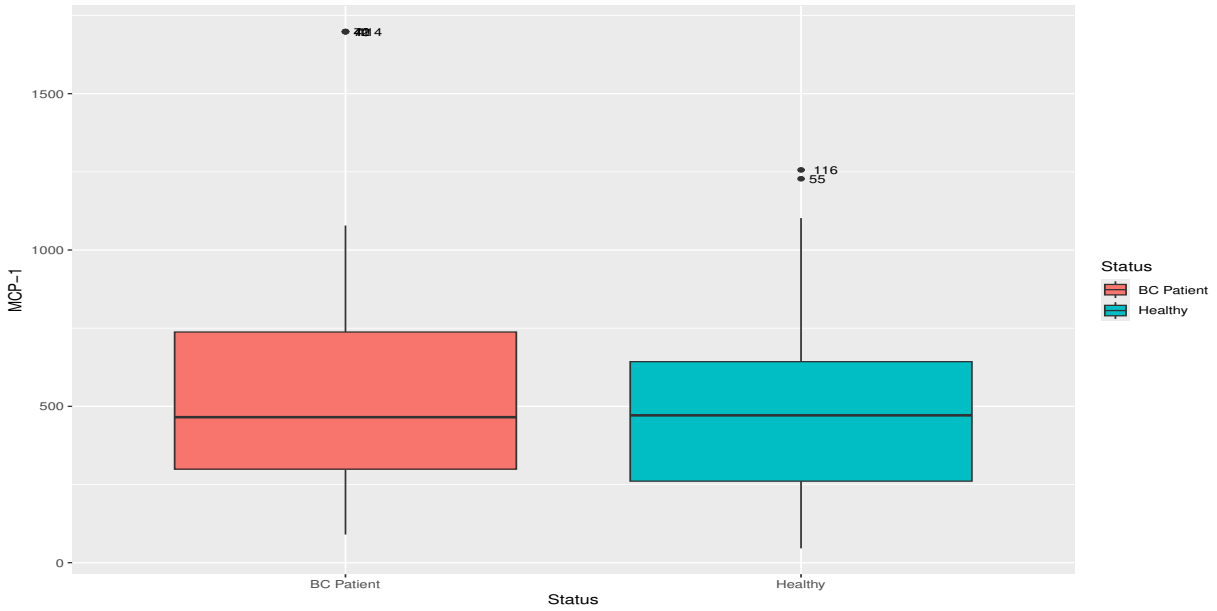

Figure 11: Box plots for MCP-1 for breast cancer and healthy subjects

as influential with observation 116 as the most influential.

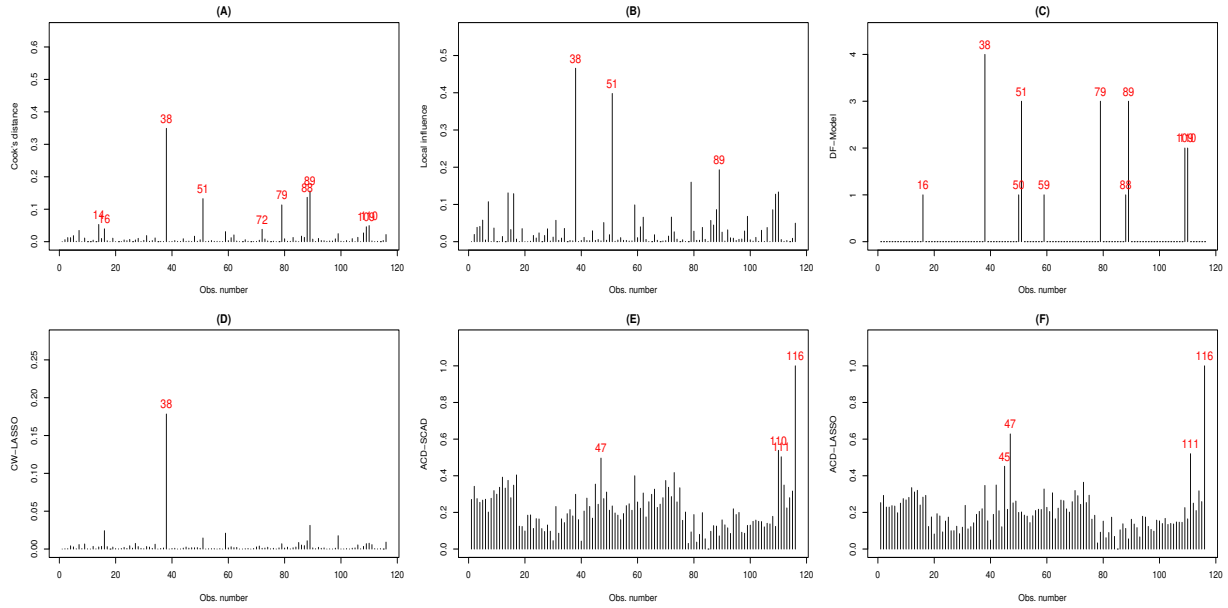

Figure 12: Influence diagnostics for MCP-1

To quantify how influence and collinearity affect model composition, we conducted a

stability experiment with the SCAD penalty. In each of 50 replicates, we drew a 95% subsample without replacement and computed SCAD estimates (i) on the raw subsample and (ii) after trimming the outliers flagged by each detection method. Table 1 reports selection proportions (bold when ≥ 0.50). BMI, Leptin, and Resistin—are repeatedly chosen before and after trimming, indicating a stable core signal. Methods employing ℓ_1 penalties (DF-MODEL, CW-LASSO) often omit HOMA index, a behavior that is consistent with the well-documented grouping tendency of LASSO under strong predictor correlation. Trimming improves selection stability for borderline variables without altering the core set.

Table 1: Proportion of selection in 50 random samples

Predictor	ALL	CKD	LOC-INF	DF-MODEL	CW-LASSO	ACD-SCAD	ACD-LASSO
Age	0.26	0.12	0.04	0.08	0.16	0.34	0.36
BMI	**1.00**	**1.00**	**1.00**	**0.98**	**1.00**	**0.94**	**0.96**
Glucose	0.18	0.04	0.20	0.38	0.34	0.30	0.40
Insulin	0.20	**0.54**	0.14	0.20	0.06	0.16	0.12
HOMA	**0.94**	0.02	**0.72**	0.22	0.36	**0.50**	**0.50**
Leptin	**1.00**	**1.00**	**0.96**	**0.84**	**0.84**	**0.74**	**0.68**
Adiponectin	0.18	0.12	0.16	0.22	0.04	0.26	0.26
Resistin	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**0.96**	**0.96**
Status[Healthy]	0.16	0.02	0.30	0.26	0.08	0.26	0.38

5.2 Application to Riboflavin Data

We evaluate all procedures—ALL, DF-MODEL, CW-LASSO, ACD-SCAD, and ACD-LASSO on a high-dimensional gene-expression dataset quantifying riboflavin (vitamin B_2) production in *Bacillus subtilis*. The dataset was brought to prominence in the high-dimensional

literature by Bühlmann et al. (2014), is distributed via the `hdi` package in `R`, and has since been reanalyzed in several studies (Zhang et al., 2019; Genç and Özkale, 2021). It comprises $n = 71$ fermentation runs and $p = 4{,}088$ predictors; each covariate is the log expression level of a gene, and the response is the log riboflavin production rate. This $p \gg n$ regime, together with pronounced gene–gene correlations, provides a stringent testbed for sparse estimation and variable selection.

Table 2: Most frequently selected genes with proportions of at least 0.5 in 50 random samples are given. Proportions below 0.5 are indicated as -.

Selected Gene	ALL	DF-MODEL	CW-LASSO	ACD-SCAD	ACD-LASSO
73 (ARGF)	0.6	-	-	-	-
624 (LYSC)	**0.6**	-	-	**0.6**	**0.6**
792 (PCKA)	0.6	-	-	-	-
1131 (SPOVAA)	0.8	-	-	-	-
1762 (YEBC)	-	-	0.6	-	-
2034 (YHDZ)	0.6	-	-	-	-
2564 (YOAB)	**0.8**	**0.6**	-	**0.6**	**0.8**
2936 (YPZD)	-	-	0.8	-	-
4003 (YXLD)	**0.8**	-	-	**0.6**	**0.6**
4004 (YXLE)	-	-	0.8	-	-

All four methods appear to flag different observations as influential although they all agree on some observation around 30 being influential. CW-LASSO and the ACD methods flagged fewer observations compared to DF-Model, which is not surprising based on previous results from the simulation study. We proceed with the stability of variable selection to quantify how influence and collinearity affect model composition.

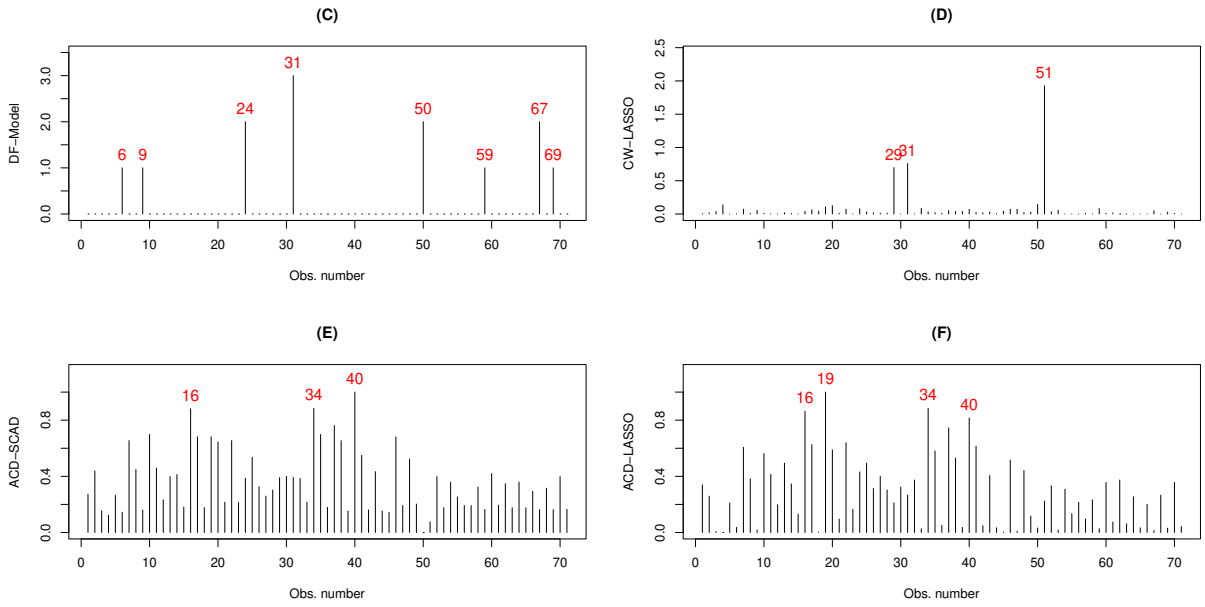

Figure 13: Influence diagnostics for Riboflavin data

Table 2 reports, for each method, the proportion of resamples in which a gene is selected; we display only genes with a selection frequency of at least 0.5 by at least one procedure. Three signals stand out for the ACD family: **LYSC (624)**, **YOAB (2564)**, and **YXLD (4003)** are each retained by ACD–LASSO and ACD–SCAD with frequencies ≥ 0.6 (YOAB reaches 0.8 under ACD–LASSO). In contrast, the DF–Model is markedly conservative: only YOAB exceeds the 0.5 threshold, while CW–LASSO emphasizes a different subset (YEBC, YPZD, YXLE at 0.6–0.8) and excludes the ACD core set. The ALL baseline, which tends to overselect, flags several genes at or above 0.6 (e.g., SPOVAA at 0.8), but these do not replicate consistently under penalized estimators. Overall, the ACD procedures concentrate their selections on a small, repeatedly chosen group, most notably LYSC, YOAB, and YXLD, indicating greater stability under resampling relative to the CW–LASSO and DF-Model.

6 Conclusion

We proposed an adaptive Cook's distance (ACD) based on sparse local gradients for influence diagnostics in high-dimensional regression models under severe collinearity and contamination. By estimating an effective direction without specifying a link and enforcing sparsity in the local estimation, ACD targets the directions along which atypical cases most perturb estimation.

In simulations that varied correlation structure and dimensionality, ACD–LASSO and ACD–SCAD reduced masking and swamping relative to case–deletion, perturbation diagnostics, and recent ℓ_1-based influence methods; trimming ACD-flagged cases yielded more stable variable selection while preserving the core signal set. The results is similar for moderate p. Two data applications underscore these gains: in the Monocyte Chemoattractant Protein-1 analysis, trimming improved the consistency of borderline covariates while maintaining a stable core; in the riboflavin genomics study, ACD concentrated on a small, reproducible subset of genes and avoided the over/under–selection seen in baselines.

We recommend pairing ACD with resampling-based stability summaries and reporting post-trim estimates alongside raw fits. Future work includes scalable implementations for ultra-high dimensions, extensions to generalized outcomes and structured penalties, and finite-sample guarantees for trimming. It would also be interesting to consider multi-index models, particularly for sufficient dimension reduction.

References

Belsley, D. A., Kuh, E., and Welsch, R. E. (1980). Regression diagnostics: Identifying influential data and sources of collinearity. *Wiley Series in Probability and Statistics*.

Bühlmann, P., Kalisch, M., and Meier, L. (2014). High-dimensional statistics with a view toward applications in biology. *Annual review of statistics and its application*, 1(1):255–278.

Chatterjee, S. and Hadi, A. S. (1986). Influential observations, high leverage points, and outliers in linear regression. *Statistical science*, pages 379–393.

Cook, R. D. (1977). Detection of influential observation in linear regression. *Technometrics*, 19(1):15–18.

Cook, R. D. (1986). Assessment of local influence. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 48(2):133–155.

Cook, R. D. and Weisberg, S. (1980). Characterizations of an empirical influence function for detecting influential cases in regression. *Technometrics*, 22(4):495–508.

Deshmane, S. L., Kremlev, S., Amini, S., and Sawaya, B. E. (2009). Monocyte chemoattractant protein-1 (mcp-1): an overview. *Journal of interferon & cytokine research*, 29(6):313–326.

Dutta, P., Sarkissyan, M., Paico, K., Wu, Y., and Vadgama, J. V. (2018). Mcp-1 is overexpressed in triple-negative breast cancers and drives cancer invasiveness and metastasis. *Breast cancer research and treatment*, 170(3):477–486.

Eaton, M. L. (1986). A characterization of spherical distributions. *Journal of Multivariate Analysis*, 20(2):272–276.

Fan, J. and Li, R. (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *Journal of the American statistical Association*, 96(456):1348–1360.

Fan, J. and Lv, J. (2008). Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 70(5):849–911.

Genç, M. and Özkale, M. R. (2021). Usage of the go estimator in high dimensional linear models. *Computational Statistics*, 36(1):217–239.

Hodges, S. and Moore, P. (1972). Data uncertainties and least squares regression. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 21(2):185–195.

Jiao, Z. and Lee, Y. (2025). Assessment of case influence in the lasso with a case-weight adjusted solution path. *Technometrics*, pages 1–14.

Kim, C., Lee, J., Yang, H., and Bae, W. (2015). Case influence diagnostics in the lasso regression. *Journal of the Korean Statistical Society*, 44(2):271–279.

Li, B., Artemiou, A., and Li, L. (2011). Principal support vector machines for linear and nonlinear sufficient dimension reduction. *The Annals of Statistics*, 39.

Li, K.-C. and Duan, N. (1989). Regression analysis under link violation. *The Annals of Statistics*, 17(3):1009–1052.

Mulholland, B. S., Hofstee, P., Millar, E. K., Bliuc, D., O'Toole, S., Forwood, M. R., and McDonald, M. M. (2023). Mcp-1 expression in breast cancer and its association with distant relapse. *Cancer Medicine*, 12(15):16221–16230.

Nurunnabi, A., Hadi, A. S., and Imon, A. (2014). Procedures for the identification of multiple influential observations in linear regression. *Journal of Applied Statistics*, 41(6):1315–1331.

Patrício, M., Pereira, J., Crisóstomo, J., Matafome, P., Gomes, M., Seiça, R., and Caramelo, F. (2018). Using resistin, glucose, age and bmi to predict the presence of breast cancer. *BMC cancer*, 18(1):29.

Peña, D. (2005). A new statistic for influence in linear regression. *Technometrics*, 47(1):1–12.

Peña, D. and Yohai, V. (1999). A fast procedure for outlier diagnostics in large regression problems. *Journal of the American Statistical Association*, 94(446):434–445.

Polasek, W. (1984). Regression diagnostics for general linear regression models. *Journal of the American Statistical Association*, 79(386):336–340.

Prendergast, L. A. (2008). Trimming influential observations for improved single-index model estimated sufficient summary plots. *Computational statistics & data analysis*, 52(12):5319–5327.

Rajaratnam, B., Roberts, S., Sparks, D., and Yu, H. (2019). Influence diagnostics for

high-dimensional lasso regression. *Journal of Computational and Graphical Statistics*, 28(4):877–890.

Roberts, S., Martin, M. A., and Zheng, L. (2015). An adaptive, automatic multiple-case deletion technique for detecting influence in regression. *Technometrics*, 57(3):408–417.

Rousseeuw, P. J. and Leroy, A. M. (2003). *Robust regression and outlier detection*. John wiley & sons.

Soale, A.-N. (2025). Detecting influential observations in single-index fréchet regression. *Technometrics*, 67(2):311–322.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288.

Wang, H. and Xia, Y. (2008). Sliced regression for dimension reduction. *Journal of the American Statistical Association*, 103(482):811–821.

Wang, Q. and Yao, W. (2012). An adaptive estimation of mave. *Journal of Multivariate Analysis*, 104(1):88–100.

Wang, T., Li, Q., Zang, Q., and Li, Z. (2019). A multiple-case deletion approach for detecting influential points in high-dimensional regression. *Communications in Statistics-Simulation and Computation*, 48(7):2065–2082.

Xia, Y. (2007). A constructive approach to the estimation of dimension reduction directions. *The Annals of Statistics*, 35(6):2654–2690.

Xia, Y., Tong, H., Li, W. K., and Zhu, L.-X. (2002). An adaptive estimation of dimension reduction space. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 64(3):363–410.

Zhang, C., Wu, Y., and Zhu, M. (2019). Pruning variable selection ensembles. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 12(3):168–184.

Zhang, C.-H. (2010). Nearly unbiased variable selection under minimax concave penalty. *The Annals of Statistics*, 38(2):894 – 942.

Zhao, J., Leng, C., Li, L., and Wang, H. (2013). High-dimensional influence measure. *The Annals of Statistics*, 41(5):2639 – 2667.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 67(2):301–320.